DMS-Eval: An Empirical Benchmark of Lightweight Object Detectors for In-Cabin Driver Monitoring

Oumar Mamoun Ibrahim
Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
0009-0008-0312-1605

Mohamad Khairi bin Ishak
Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
0000-0002-3554-0061

***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—driver monitoring systems, driver drowsiness detection, driver distraction detection, lightweight object detection, benchmark evaluation, computer vision, YOLO, DETR, real-time edge AI**

I. Introduction

In-cabin Driver Monitoring Systems (DMS) constitute a vital component of modern Advanced Driver Assistance Systems (ADAS) and autonomous vehicle safety suites, aiming to prevent roadway collisions through early visual detection of driver drowsiness and inattention. While high-level driver state assessment often leverages temporal recurrent or multi-frame sequence models, automotive edge deployments impose severe compute, memory, and latency constraints. Consequently, there is an urgent practical demand for lightweight, nano-scale 2D object detectors capable of executing real-time single-frame inference on embedded in-cabin hardware.

To systematically evaluate modern compact detector paradigms under strictly controlled single-frame constraints, this paper introduces **DMS-Eval**, an empirical benchmark comparing real-time nano-scale object detectors. We evaluate leading single-stage convolutional detectors (Ultralytics YOLO11n [12] and Ultralytics YOLO26n [13]) against modern end-to-end Real-Time Detection Transformers (D-FINE-N [14]).

The benchmark is developed on real-cabin RGB video from the Driver Monitoring Dataset (DMD) [15], incorporating all three original dataset session categories: *distraction*, *drowsiness*, and *gaze*. Across 81 driver-facing videos from 14 volunteer participants, frames are sampled uniformly at 1 FPS and cropped to a frozen 640×640 spatial window ($x = 272, y = 71, w = 640, h = 640$) as illustrated in Fig. 1, yielding a standardized dataset of 15,723 frames. Retaining all three session folders—particularly the *gaze* recordings and non-cue intervals—provides an abundant volume of naturalistic negative background frames (0 bounding boxes), preventing compact detectors from overtraining on positive targets and enforcing realistic false-positive suppression during routine driving. Ground truth is constructed through 100% direct manual expert annotation in Label Studio [16] with format exports structured via `label-studio-converter` [17] targeting 4 frozen visual warning cues (exemplified in Fig. 2):

1) `yawning`: mouth region only (wide oral opening/distension);
2) `hand_over_mouth`: full head/face enclosing the occluding hand;
3) `drinking`: driver face and beverage container together in active consumption posture;
4) `phone_use`: handheld phone and interacting hand held to ear/head in calling posture.

Under our single-frame annotation protocol, each sampled image contains at most one visual warning cue annotation (0 or 1 bounding box per image). In particular, `yawning` and `hand_over_mouth` are mutually exclusive and are never labeled twice in the same image; if a driver yawns while a hand visibly covers or occludes the mouth, the instance is uniquely labeled as `hand_over_mouth` (full head/face), ensuring clear, non-overlapping ground-truth targets.

Under the controlled-comparison principle, all models are evaluated using identical data partitions under a strict 8 / 3 / 3 subject-disjoint split (8 Train, 3 Validation, 3 Test subjects) to prevent identity leakage, utilizing identical hardware (NVIDIA RTX 4060, 8 GB VRAM), batch size 1, 220 fine-tuning epochs without early stopping, fixed random seed (13), and FP16 automatic mixed precision. Optimal checkpoints and confidence thresholds (τ) are selected strictly on the validation split prior to an isolated, single-pass test evaluation.

A. Research Question and Scope

This study is limited to **frame-level observable cues of driver drowsiness and distraction**, rather than temporal driver-state inference. Head orientation behaviors (e.g., turning away or looking toward side and rearview mirrors) are intentionally excluded from the static 2D detection ontology. Drivers routinely perform frequent mirror checks and visual scanning as part of safe driving; in isolated 1 FPS static frames without continuous temporal video context or 3D gaze tracking, there is no consistent or objective boundary to distinguish safe, brief mirror glances from dangerous, prolonged inattention. To eliminate subjective label noise from split datasets, the benchmark focuses strictly on the four unambiguous, self-contained visual warning cues.

> **RQ:** *Under a controlled, compute-constrained evaluation, how do lightweight object detectors compare in detection accuracy, inference efficiency, and robustness when identifying visual cues associated with driver distraction and drowsiness under normal and low-light/nighttime driving conditions?*

II. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections II-A to II-H below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable. See Table I for the primary abbreviations and acronyms used throughout the DMS-Eval benchmark.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not "Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

D. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited,

such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)

- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [7].

F. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

G. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

H. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. **??**", even at the beginning of a sentence.

TABLE I
LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Definition / Description
DMS	Driver Monitoring System
DMD	Driver Monitoring Dataset
YOLO	You Only Look Once
DETR	DEtection TRansformer
mAP	Mean Average Precision
FPS	Frames Per Second
FLOPs	Floating Point Operations
GFLOPs	Giga Floating Point Operations (10^9 FLOPs)
NMS	Non-Maximum Suppression
EAR	Eye Aspect Ratio
COCO	Common Objects in Context
IoU	Intersection over Union
AMP	Automatic Mixed Precision (FP16)
VRAM	Video Random Access Memory

b) Benchmark Status and Pending Updates: The following sections document benchmark components that are frozen alongside components awaiting empirical execution:

- **Dataset Partitions**: 14 subjects with 8/3/3 subject-disjoint split. `[TO BE UPDATED: Exact Train, Validation, and Test Subject ID allocation in splits.json]`.
- **Sampling Implementation**: Uniform 1 FPS sampling policy. `[TO BE UPDATED: Exact mapping rule from` $\approx$ 29.76 `FPS source video to 1 FPS extracted frames]`.
- **Software Environment**: NVIDIA RTX 4060 (8 GB VRAM). `[TO BE UPDATED: Exact pinned software versions for CUDA, PyTorch, cuDNN, Ultralytics, D-FINE commit, and THOP]`.
- **Validation Calibration**: Shared F1-maximization threshold sweep. `[TO BE UPDATED: Validation-selected optimal confidence thresholds` $(\tau_{\mathrm{YOLO11n}}, \tau_{\mathrm{D\text{-}FINE\text{-}N}}, \tau_{\mathrm{YOLO26n}})$`]`.
- **Detection Benchmark Results**: Shared evaluator ($mAP@0.5:0.95$, $mAP@0.5$, Precision, Recall, F1). `[TO BE UPDATED: Empirical test-set detection performance metrics across candidate models]`.
- **Inference and Footprint Metrics**: FP16 median latency, throughput, THOP GFLOPs, disk size. `[TO BE UPDATED: Profiling measurements on target RTX 4060 GPU]`.

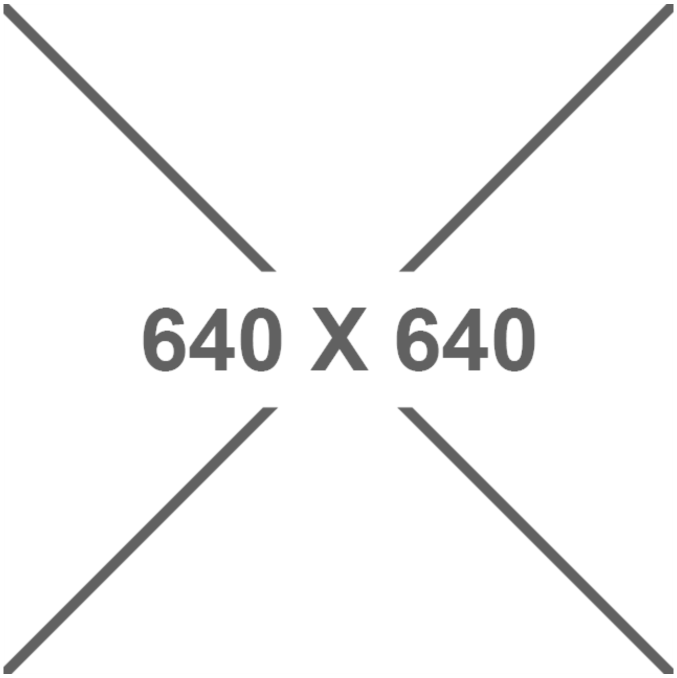

Fig. 1. In-cabin driver-facing frame extraction on DMD video with standardized 640×640 spatial window cropping ($x = 272, y = 71, w = 640, h = 640$), centering the driver's head, face, and torso for low-latency single-frame object detection.

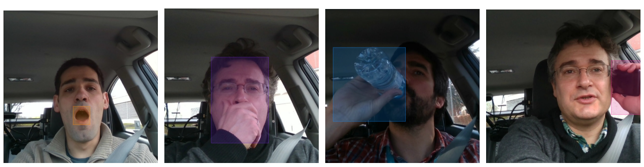

Fig. 2. Visual demonstrations of our manual annotations in Label Studio across all 4 frozen target warning cues: (a) `yawning` (orange bounding box surrounding mouth aperture only), (b) `hand_over_mouth` (purple bounding box enclosing full head, face, and occluding hand), (c) `drinking` (blue bounding box enclosing face and beverage container in active consumption posture), and (d) `phone_use` (pink bounding box enclosing hand and handheld device held at the ear in calling posture).

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [4]. Papers that have been accepted for publication should be cited as "in press" [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

References

[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, "Title of paper if known," unpublished.

[5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

[8] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, arXiv:1312.6114. [Online]. Available: https://arxiv.org/abs/1312.6114

[9] S. Liu, "Wi-Fi Energy Detection Testbed (12MTC)," 2023, gitHub repository. [Online]. Available: https://github.com/liustone99/Wi-Fi-Energy-Detection-Testbed-12MTC

[10] "Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009." U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2

[11] K. Eves and J. Valasek, "Adaptive control for singularly perturbed systems examples," Code Ocean, Aug. 2023. [Online]. Available: https://codeocean.com/capsule/4989235/tree

[12] G. Jocher and J. Qiu, "Ultralytics YOLO11," 2024. [Online]. Available: https://github.com/ultralytics/ultralytics

[13] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO26: End-to-End Real-Time Object Detection," 2026. [Online]. Available: https://github.com/ultralytics/ultralytics

[14] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-FINE: Redefine Regression Task in DETRs as Fine-grained Distribution Refinement," arXiv preprint arXiv:2410.13842, 2024.

[15] J. D. Ortega *et al.*, "DMD: A Large-Scale Multi-modal Driver Monitoring Dataset for Attention and Alertness Analysis," in *Computer Vision – ECCV 2020 Workshops*, Springer, 2020, pp. 387–405.

[16] HumanSignal, "Label Studio: Open Source Data Labeling Tool," 2024. [Online]. Available: https://github.com/HumanSignal/label-studio

[17] HumanSignal, "Label Studio Converter: Format Exporters and Converters for Label Studio Annotations," 2024. [Online]. Available: https://github.com/HumanSignal/label-studio-converter

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove the template text from your paper may result in your paper not being published.